

Sophia N. Wilson

☎ (+45) 24 22 25 32
✉ sophia.wilson@di.ku.dk
📁 [sophiawilson18.github.io](https://github.com/sophiawilson18)
🐙 Github in LinkedIn
🔍 Google Scholar



Biography

Sophia N. Wilson is a Ph.D. fellow at the Sustainable Artificial Intelligence for Sciences (SAINTS) Lab, Dept. of Computer Science, University of Copenhagen (UCPH), supervised by Raghavendra Selvan and Erik Dam. Her research spans two strands: a technical strand developing algorithms and AI systems that are more resource efficient in data, compute, and energy use; and a socio-technical strand measuring and mitigating sustainability harms in the AI industry, including environmental impact and poor labour conditions. She is also active in bridging academia and industry, and in outreach promoting STEM fields, particularly to girls and young women.

Education

- 2025–present **PhD Fellow, Sustainable AI**, SAINTS Lab, Dept. of Computer Science, UCPH, Denmark.
Thesis: Glean: Towards Green Lean Data in the Era of Big Dark Data.
Supervision: Raghavendra Selvan and Erik Dam.
- 2023–2025 **MSc in Computational Physics**, Niels Bohr Institute, UCPH, Denmark.
Thesis: "Quantifying the Reduction in Carbon Footprint of Physics-Informed Machine Learning".
Supervision: Raghavendra Selvan and Jens Hesselbjerg Christensen.
- 2019–2023 **BSc in Physics**, Niels Bohr Institute, UCPH, Denmark.
Thesis: "Measuring Cosmological Parameters with Fast Radio Bursts".
Supervision: Kasper Heintz and Johan Fynbo.

Selected Publications

In Conference Proceedings

- 2026 **Sophia N. Wilson**, Andrew Millard, Guðrún Fjóla Guðmundsdóttir, Raghavendra Selvan, and Sebastian Mair. Position: Stop Preaching and Start Practising Data Frugality for Responsible Development of AI. In *Proceedings of the 43rd International Conference on Machine Learning (ICML), Position Paper Track*, 2026.
- 2026 **Sophia N. Wilson**, Sebastian Mair, Mophat Okinyi, Erik B Dam, Janin Koch, and Raghavendra Selvan. How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026.
- 2026 **Sophia N. Wilson**, Jens Hesselbjerg Christensen, and Raghavendra Selvan. Performance-Carbon Trade-Offs across Architectural Biases in Shear Flow Forecasting. In *Proceedings of the 19th European Conference on Computer Vision (ECCV), LIMIT Workshop*, 2026.

Preprints

- 2026 Pedram Bakhtiarifard, **Sophia N. Wilson**, Mahmoud Afifi, Jonathan Wenshøj, and Raghavendra Selvan. Characterizing Learning in Deep Neural Networks using Tractable Algorithmic Complexity Analysis, 2026.

International Experience

- Aug. 2026 **Cetinje, Montenegro**, Eastern European Machine Learning Summer School (Best Poster).
- Feb. 2026 **Melbourne, Australia**, Machine Learning Summer School.
- Spring 2024 **University of Amsterdam, The Netherlands**, Exchange Semester.
- Aug. 2022 **University of Montréal, Canada**, Astromatic Machine Learning Summer School.
- Aug. 2022 **Nordic Optical Telescope, La Palma, Spain**, Summer School.

Teaching

- 2026 **Data Analysis and Machine Learning**, MSc course for high school teachers, Dept. of Computer Science, Aarhus University.
- 2025 **Machine Learning A**, MSc course, Dept. of Computer Science, UCPH.
- 2025 **Mathematics A**, high school, Falkonergårdens Gymnasium.
- 2023 **Summer Science Camp**, elementary school, Insero.
- 2022-23 **Mathematics for Physicists 2**, BSc course, Niels Bohr Institute, UCPH.
- 2022-23 **Mathematics for Physicists 1**, BSc course, Niels Bohr Institute, UCPH.
- 2020-21 **Mechanics and Special Relativity**, BSc course, Niels Bohr Institute, UCPH.

Supervision

- 2026 **Co-supervised two BSc students (with R. Selvan)**, "The Environmental Impact of Generative AI: A Life Cycle Assessment as a Basis for Studying Consumer Perception", Dept. of Communication, UCPH.

Talks

- Sep. 2026 **High Energy Physics Software Foundation Seminars**, Webinar, "Assessing and Mitigating Sustainable Costs of Frontier AI".
- Sep. 2026 **Center for AI, Danish School of Media and Journalism**, Webinar, "Bæredygtig AI: Udfordringer og Løsninger".
- May 2026 **Pioneer Centre for AI, UCPH**, Workshop on Green AI, presenting "How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI".
- May 2026 **National Center for AI in Society, UCPH**, Workshop on Sustainable AI, "On the Challenges in Assessing the Sustainability of AI" (joint with R. Selvan).
- May 2026 **Technical University of Denmark, Visual Computing Group**, Presenting "How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI".
- Apr. 2026 **Technical University of Denmark, Embedded AI Group**, Presenting "Stop Preaching and Start Practising Data Frugality for Responsible Development of AI".
- Apr. 2026 **Lund Stem Cell Center, Lund University**, Workshop on Sustainable AI, "How to measure and manage energy consumption of AI models".
- Aug. 2025 **D3A**, Workshop on Resource-Aware Machine Learning, "Quantifying the Reduction in Carbon Footprint of Physics-Informed Machine Learning".
- Apr. 2025 **Copenhagen Center for Social Data Science, UCPH**, "On the Carbon Footprint of AI: From Monitoring to Mitigation" (joint with R. Selvan).
- Mar. 2025 **Niels Bohr Institute, UCPH**, 4th Annual NBI MSc Student Symposium, "Quantifying the Reduction in Carbon Footprint of Physics-Informed Machine Learning".

Panels

- Oct. 2026 **ACM Conference on AI, Ethics, and Society, Student Symposium**, "Responsible AI".

- Jun. 2026 **Folkemødet, Bornholm**, "Kærlighed, Klimakrise og Kunstig Intelligens".
- Apr. 2026 **Lund Stem Cell Center, Lund University**, "Sustainable AI in Practice: From Computational Costs to Real-World Impacts".

Guest Lecturing

- Feb. 2026 **University College Copenhagen**, BSc course on Ethics and Technology, "AI, Resource Use, and Sustainability".
- May 2025 **Dept. of Computer Science, UCPH**, PhD course on Machine Learning for Sciences, "Physics-Informed Machine Learning as a Carbon-Efficient Approach".

Academic Service

- Sep. 2026 **Programme Committee Member**, AAAI 2026 Fall Symposium on Parsimonious Scientific Machine Learning: Accelerating Scientific Discovery with AI under Data and Compute Constraints.
- Sep. 2026 **Admissions Committee Member**, BSc in Machine Learning (quota 2), Dept. of Computer Science, UCPH.

Organisation

- Mar. 2026 **SAINTS Lab, Dept. of Computer Science, UCPH**, Co-organiser and moderator, SAINTS'26 Workshop on Sustainable AI for Sciences.

Outreach

- Sep. 2025 **Det Åbne Gymnasium**, "Energiforbrug og CO₂-aftryk af kunstig intelligens".
- May 2025 **Falkonergårdens Gymnasium**, "Hvordan er det at læse fysik på universitetet?".
- 2021-24 **Women in Physics**, Contributing to initiatives promoting gender balance in STEM.

Media

- Sep. 2026 **Podcast interview by Verden Forsøgt Forklaret**.
<https://open.spotify.com/episode/7wsAAKJJJaPAr6avWYEUfF3?si=dee33aab14d1460b&nd=1&dlsi=1c3cff7934644ef7>
- Sep. 2026 **Interview by Uniavisen**.
<https://uniavisen.dk/forsker-ai-bliver-aldrig-en-baeredygtig-teknologi/>
- Dec. 2025 **Interview by Women in Physics**.
<https://kvinderifysik.dk/2025/12/15/interview-with-sophia-wilson/>